

# Signals and Systems

1. Signal – Function representing a physical quantity over time.
2. System – Entity that processes input signal to produce output.
3. Continuous Signal – Defined for all time values.
4. Discrete Signal – Defined only at specific time intervals.
5. Analog Signal – Continuous in time and amplitude.
6. Digital Signal – Discrete in time and amplitude.
7. Periodic Signal – Repeats after a fixed interval.
8. Aperiodic Signal – Does not repeat.
9. Even Signal – Symmetric about vertical axis.
10. Odd Signal – Antisymmetric about origin.
11. Unit Step Signal – Signal that jumps from 0 to 1 at  $t=0$ .
12. Unit Impulse Signal – Instantaneous spike at  $t=0$ .
13. Ramp Signal – Linearly increasing signal with time.
14. Time Shifting – Delaying or advancing a signal.
15. Time Scaling – Compressing or expanding signal in time.
16. Linearity – Follows superposition principle.
17. Time Invariance – System properties do not change over time.
18. Causality – Output depends only on present and past inputs.
19. Stability (BIBO) – Bounded input gives bounded output.
20. Convolution – Operation to find system output.
21. Fourier Series – Represents periodic signals as sum of sinusoids.
22. Fourier Transform – Converts signal to frequency domain.
23. Laplace Transform – Generalized frequency domain representation.
24. Z-Transform – Analysis of discrete-time systems.
25. Frequency Response – System behavior vs frequency.
26. Bandwidth – Range of frequencies a system passes.
27. Sampling – Converting continuous signal to discrete.
28. Aliasing – Overlap due to under-sampling.
29. Nyquist Rate – Minimum sampling rate ( $2 \times$  max frequency).
30. Energy Signal – Finite energy over time.